Course Intro and Random Variables

Math 742

1/12/2026

Intro

This course lies at the heart of your training as a data scientist and researcher. In every area of data science, from designing controlled experiments to building machine learning models and developing new inference methods. Statistical reasoning is the foundation. In this course, we will build the language and logic that allow you to understand uncertainty rigorously and justify your conclusions with data.

Mathematical statistics is not just a collection of formulas; it is a framework for thinking. It unifies probability theory, estimation, decision-making, and asymptotic reasoning into a coherent whole. By the end of the semester, you won't simply apply statistical methods, you'll be able to derive them, analyze their properties, and defend your modeling choices mathematically. That depth of understanding is what distinguishes a PhD trained data scientist from someone who merely runs code.

Why It Matters for You:

- Nearly every area in modern data science: Bayesian inference, causal discovery, high-dimensional modeling, and even deep learning, rests on statistical principles. Questions like "What can we infer from the data?" or "How confident are we in our conclusions?" are inherently statistical and require detailed statistical knowledge to answer. Whether you go on to work in academia, industry research, or applied analytics, your ability to reason precisely about randomness and uncertainty will shape how others trust your results.

The exercises (homework, labs, proofs, and simulations) may feel abstract at first, but they build the conceptual muscles that makes advanced methods second nature later. You may find the material from this course will help you draft research papers, grant proposals, and even assist in conversations with collaborators who want to know why a method works.

Setting Expectations:

This course will ask you to engage deeply, derive results, question assumptions, and connect the theory to computation. But it will also give you a powerful toolkit. Once you understand the theoretical machinery of inference, you can build your own methods and not just use someone else's. By the end of the course, you should be able to move seamlessly between theory and practice. Mathematical statistics is the common language of scientific discovery in the data age. Master it here, and you'll carry that fluency throughout your PhD and career.

Basic Probability

Definition 1: The set of all possible outcomes of an experiment is called the **sample space**.

Example 1: Roll a pair of 6-sided dice. List the elements of the sample space using (x, y) where x is the number of dots face up on the first die and y is the number of dots face up on the second die.

$S =$ {(1,1) (1,2) (1,3) (1,4) (1,5) (1,6) (2,1) (2,2)... (6,4) (6,5) (6,6) }

Definition 2: An **event** is a collection of outcomes from the sample space.

Example 2: In example 1 we rolled 2 six-sided dice. Let event A be the outcomes where the total number of dots face up (i.e. die1 + die2) is 7.

A ={(1,6) (2,5) (3,4) (4,3) (5,2) (6,1)}

Random Variables

In many experiments it's easier to deal with a summary variable rather than the outcomes themselves. Suppose a "balanced" (i.e. Pr(H)=Pr(T)) coin is flipped 50 times. Imagine if we wanted to write down the entire sample space. For one outcome, we would have to write down a sequence of H and T's that has length 50. In total, there would be 2^{50} of such sequences which is $\approx 1.13 \times 10^{15}$!

Instead of thinking about the outcomes themselves, in most instances it is sufficient just to know how many heads occurred. Let X be the number of heads in 50 flips. Now, the sample space for heads only has 51 elements in it instead of 2^{50} elements.

$X = \{0, 1, 2, ..., 49, 50\}$

From this you can see that we have substantially reduced the number of things we have to keep track of. Mathematically, X is a mapping function that maps the actual outcome to an integer 0 to 50. Notice, there is no requirement for the function to be 1-1. Meaning, two outcomes can map to the same integer. In probability theory, the variable X is referred to as a **random variable**. It's value is random and it varies each time your run the experiment.

Definition 3: A random variable is a function that maps an outcome from the sample space S into the real numbers.

Definition 4: A function $\mathbb{P}$ that assigns a real number $\mathbb{P}(A)$ to each event A is a **probability distribution** or **probability measure** if it satisfies the following three axioms:

- Axiom 1: $\mathbb{P}(A) \geq 0$ for every event A.
- Axiom 2: $\mathbb{P}(S) = 1$
- Axiom 3: If $A_1, A_2, ...$ are disjoint events (i.e. events that cannot occur simultaneously), then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

These 3 axioms are attributed to Andrey Kolmogorov[1].

Example: Suppose we flip a balanced coin. The sample space, S ={H,T}. Define a probability measure $\mathbb{P}$ on S as follows. If the coin is balanced, $\mathbb{P}(\mathrm{H}) = \mathbb{P}(\mathrm{T}) = 0.5$.

- Defined in this way, both probabilites are ≥ 0 so axiom 1 is satisfied.
- $\mathbb{P}(S) = \mathbb{P}(\mathrm{H}) + \mathbb{P}(\mathrm{T}) = 0.5 + 0.5 = 1$, so axiom 2 is satisfied.
- If $A_1 = \{\mathrm{H}\}$ and $A_2 = \{\mathrm{T}\}$, $A_1 \cup A_2 = \{\mathrm{H, T}\} = S$. This means that $\mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2)$. Therefore, axiom 3 is satisfied.

We now introduce the **cumulative distribution function** or CDF.

[1] Kolmogorov, A. N. (1933). Grundbegriffe der Wahrscheinlichkeitsrechnung (Foundations of the Theory of Probability).

Definition 5: The **cumulative distribution function** or CDF is a function that maps the real numbers into the interval [0,1]. $F_X : \mathbb{R} \to [0,1]$

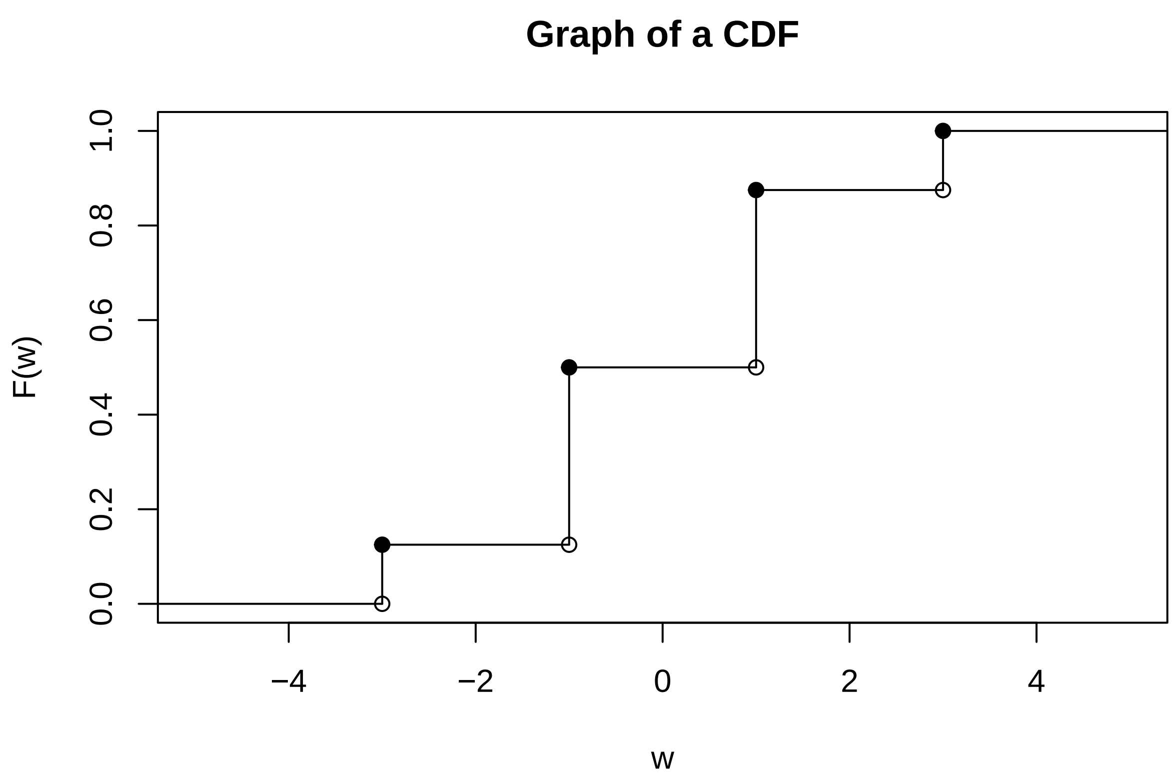
defined by,

$$F_X(x) = \Pr(X \le x)$$

Example: Suppose a balanced coin is flipped 3 times. Let W be the random variable that indicates the # Heads$-$# Tails. The sample space S ={HHH,TTT,HHT,HTH,THH,TTH,THT,HTT}, W ={-3,-1,1,3}}. Thus, $\mathbb{P}(w=-3) = 1/8$,$\mathbb{P}(w=-1) = 3/8$, $\mathbb{P}(w=1) = 3/8$, $\mathbb{P}(w=3) = 1/8$ The CDF would be:

$$F_W(w) = \begin{cases} 0 & w < -3 \\ \frac{1}{8} & -3 \le w < -1 \\ \frac{4}{8} & -1 \le w < 1 \\ \frac{7}{8} & 1 \le w < 3 \\ 1 & w \ge 3 \end{cases}$$

The graph of the CDF is a piecewise function.



Make sure you understand this. CDFs can be a little confusing. Whereas, W can only take on a finite set of values, the range of the CDF is all real numbers. Notice that, for example, $F(2) = 7/8$, but 2 is not a possible value of W.

Theorem: A function F mapping the real line to the interval $(0,1)$ is a CDF for some probability measure $\mathbb{P}$ if and only if F satisfies the following 3 conditions:

1. F is non-decreasing: $x_1 < x_2$ implies that $F(x_1) \le F(x_2)$.
2. F is normalized, which means:

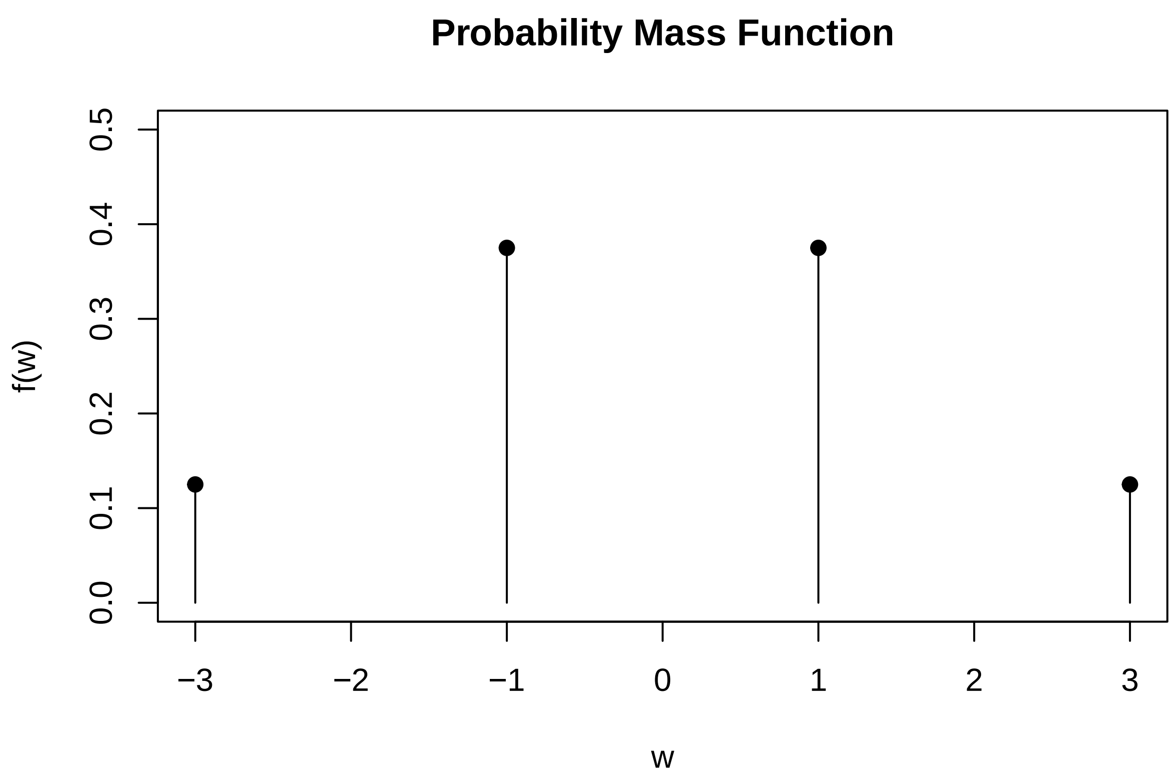
- $\lim_{x \to -\infty} F(x) = 0$
- $\lim_{x \to +\infty} F(x) = 1$

3. $F(x)$ is continuous from the right for all values of x.

These 3 conditions characterize a CDF uniquely (i.e., *Necessary and sufficient conditions*).

Definition 6: If X is a **discrete** random variable, it takes countably many values: $\{x_1, x_2, ...\}$. The **probability mass function**, $f_x(x) = \mathbb{P}(X = x)$. Note: x is a specific value of the random variable X.

Example: The graph of the PMF ($f(w)$) for the previous example would be:



Definition 7: A random variable X is **continuous** if there exists a function f_X such that $f_X(x) \geq 0$ for all x, $\int_{-\infty}^{\infty} f_X(x)dx = 1$ and for every $a \leq b$ $\mathbb{P}(a < X < b) = \int_a^b f_X(x)dx$.

$f_X(x)$ is called the **probability density function** (PDF) and the associated CDF would be:

$$F_X(x) = \int_{-\infty}^{x} f_X(s)ds$$

Notice that, by the fundamental theorem of calculus, $f_X(x) = F'_X(x)$ at all points x for which F_X is differentiable.

Example: Consider the random variable X whose PDF is given by,

$$f_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}e^{-x/2} & x \geq 0 \end{cases}$$

First we verify that $f_X(x)$ is indeed a pdf.

$$\int_{-\infty}^{\infty} f_X(x)dx = \int_0^{\infty} f_X(x)dx$$
$$= \int_0^{\infty} \frac{1}{2}e^{-x/2}dx$$
$$= \lim_{a\to\infty} \int_0^{a} \frac{1}{2}e^{-x/2}dx$$
$$= \lim_{a\to\infty} \left(-e^{-x/2}\right)\Big|_0^a = \lim_{a\to\infty} \left(-e^{-a/2}\right) - (-e^0) = 1$$

The CDF would be,

$$F_X(x) = \int_0^{x} \frac{1}{2}e^{-s/2}ds$$
$$= \left(-e^{-s/2}\right)\Big|_0^x = 1 - e^{-x/2}$$
$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x/2} & x \geq 0 \end{cases}$$

Definition 8: Let X be a random variable with CDF $F_X(x)$. The **inverse CDF** or **quantile** function is defined by,

$$F^{-1}(q) = \inf\{x : F_X(x) > q\}$$

Notice that if F is increasing and continuous, then if $x = F^{-1}(q)$ it is the unique real number so that $F(x) = q$.

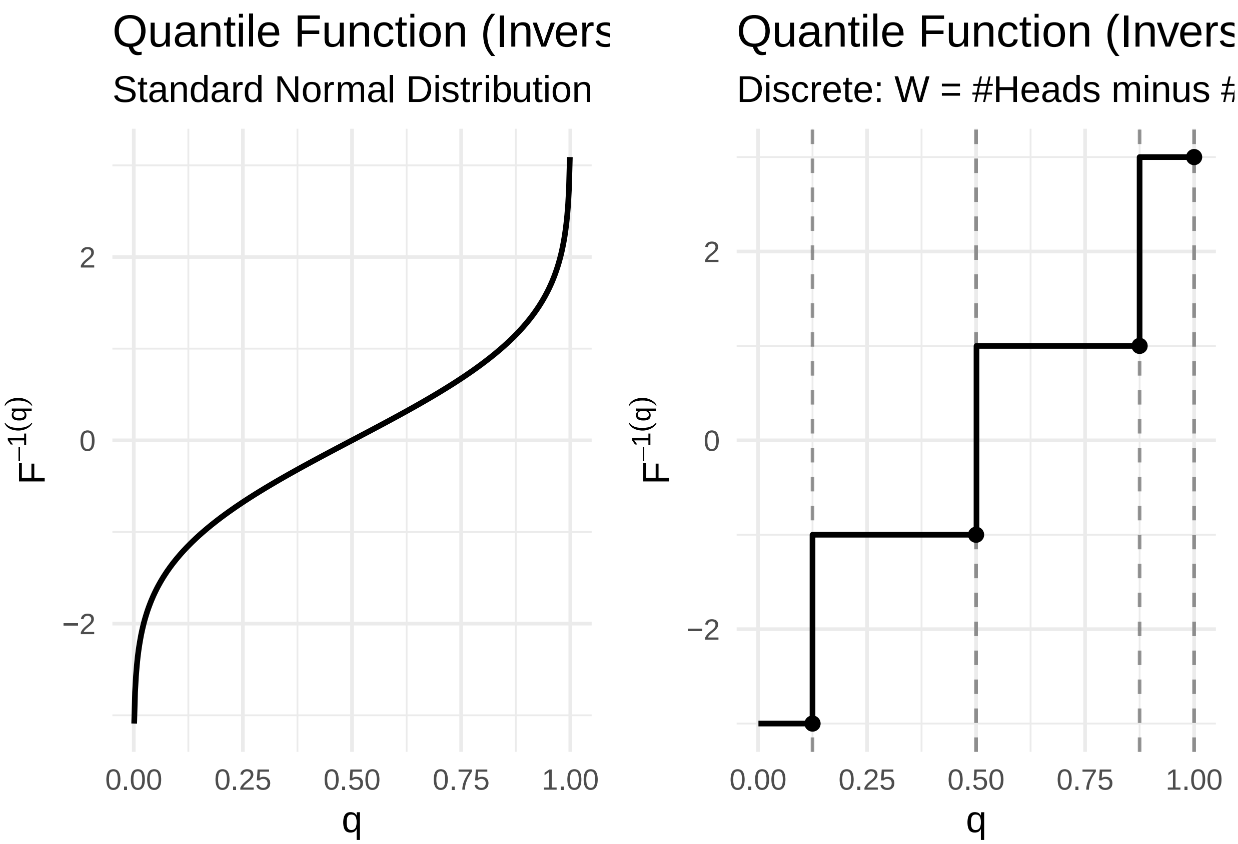
The figure on the top of the next page illustrates the quantile function $F^{-1}(q)$, or inverse CDF, for both a continuous and a discrete random variable.

Left Plot (Continuous case, Normal Distribution)

- $F^{-1}(q)$ is a smooth continuous curve
- As $q \to 0$, the quantile goes to $-\infty$.
- As $q \to 1$, the quantile goes to $+\infty$.
- This reflects the fact that the Normal distribution has no jumps and infinite support.

Right Plot (Discrete case: $W = \#$ Heads $- \#$ Tails)

- The quantile function is a step function.
- It stays at $-3, -1, 1,$ or 3 over intervals of q, then jumps upward at the cumulative probability values of $1/8, 4/8, 7/8$ and 1.
- The vertical dashed lines mark the locations of the CDF jumps.



Definition 9: The **expected value** (i.e. mean or first moment) of X is defined as:

$$\mathbb{E}(X) = \int x dF(x) = \begin{cases} \sum_x x f(x) & X \text{ is discrete} \\ \int x f(x) dx & X \text{ is continuous} \end{cases}$$

One can think of the expected value as a weighted average where each x is weighted by its probability.

Note[2] : $\int x dF(x)$ is used as short hand notation so that we don't have to write either $\sum_x x f(x)$ when X is discrete and $\int x f(x) dx$ when X is continuous.

In order to make sure that $\mathbb{E}(x)$ is defined, we say that $\mathbb{E}(x)$ is defined if $\int x dF(x) < \infty$, otherwise it does not exist.

Example: Consider the discrete random variable X whose PMF is given by,

$$f(x) = \frac{\binom{3}{x}\binom{2}{2-x}}{\binom{5}{2}}; x = 0, 1, 2, 3$$

The PMF above is the probability function for the **Hypergeometric** distribution. You can think about it as a box containing 5 balls, 3 are red and 2 are blue. 2 balls are picked without replacement, X is the number of red balls chosen. The terms in the PMF are called binomial coefficients.

$$\binom{N}{k} = \frac{N!}{(N-k)!k!}$$

[2]This is a Riemann-Stieltjes integral, which unifies discrete and continuous cases.

where, the ! is called a **factorial** $N! = N \times (N-1) \times ... \times 3 \times 2 \times 1$

$$\mathbb{E}(x) = \sum_{x=0}^{2} x f(x)$$

$$f(0) = \frac{1 \times 1}{\binom{5}{2}} = \frac{1}{10}$$

$$f(1) = \frac{3 \times 2}{\binom{5}{2}} = \frac{6}{10}$$

$$f(2) = \frac{3 \times 1}{\binom{5}{2}} = \frac{3}{10}$$

$$\mathbb{E}(x) = 0\frac{1}{10} + 1\frac{6}{10} + 2\frac{3}{10} = \frac{12}{10} = 1.2$$

Example: Consider the continuous random variable T whose PDF is given by,

$$f_T(t) = \begin{cases} 2t & 0 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\mathbb{E}(T) = \int_0^1 t2tdt = \int_0^1 2t^2 dt$$

$$= \left.\frac{2t^3}{3}\right|_0^1 = \frac{2}{3}$$

Theorem: Let the random variable X have PDF $f_X(x)$ and let a be a constant. Then, $\mathbb{E}(aX) = a\mathbb{E}(X)$.

Theorem: If $X_1, X_2, ..., X_n$ are random variables and $a_1, a_2, ..., a_n$ constants, then

$$\mathbb{E}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i \mathbb{E}(X_i)$$

Definition 10: The **variance** of a discrete random variable X is given by,

$$\sigma_X^2 = \mathbb{V}(x) = \mathbb{E}(x-\mu)^2 = \sum_{\text{all } X} (x-\mu)^2 f_x(x)$$

if X is discrete and by,

$$\sigma_X^2 = \mathbb{V}(x) = \mathbb{E}(x-\mu)^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f_x(x) dx$$

if X is continuous. Note:$\mu = \mathbb{E}(X)$.

Theorem: $\sigma_X^2 = \mathbb{V}(x) = \mathbb{E}(X^2) - \mu^2$

Proof: I will assume X is discrete. If X is continuous replace all of the summation symbols with integrals.

$$\mathbb{V}(X) = \sum_{i=1}^{n} (x-\mu)^2 f_X(x)$$

$$= \sum_{i=1}^{n} (x^2 - 2x\mu + \mu^2) f_X(x)$$

$$= \sum_{i=1}^{n} (x^2 f_X(x) - 2x\mu f_X(x) + \mu^2 f_X(x))$$

$$= \sum_{i=1}^{n} x^2 f_X(x) - 2\mu \sum_{i=1}^{n} x f_X(x) + \mu^2 \sum_{i=1}^{n} f_X(x)$$

The first sum is $\mathbb{E}(X^2)$, the second sum is $\mathbb{E}(X) = \mu$ and the final sum is 1.

Putting it all together,

$$= \mathbb{E}(X^2) - 2\mu^2 + \mu^2 = \mathbb{E}(X^2) - \mu^2$$

Theorem: For any random variable X and constants a and b, the variance of $aX + b$ is equal to $a^2\mathbb{V}(X)$.

Note: This theorem is generalizable to any number of random variables and constants.

Example: Suppose a coin is flipped until the first head. Let X be the flip number where the first head was achieved. Further, assume that the coin may not be balanced so that $\mathbb{P}(\mathrm{H}) = p$ where $(0 \leq p \leq 1)$. Determine the $\mathbb{E}(X)$ and $\mathbb{V}(X)$.

It is easy to show the $f_X(x) = (1-p)^{x-1}p$. Note: This is the PMF for the **geometric** distribution.

$$\mathbb{E}(X) = \sum_{x=1}^{\infty} x(1-p)^{x-1}p = p\sum_{x=1}^{\infty} x(1-p)^{x-1}$$

So, we have p times an infinite series. It turns out this infinite series is the series you get when you differentiate a geometric series with $a = 1$.

$$\sum_{x=0}^{\infty} r^x = \frac{1}{1-r}, |r| < 1$$

The derivative of the series on the left side converges to the derivative of the right hand side.

$$\sum_{x=1}^{\infty} xr^{x-1} = \frac{1}{(1-r)^2}, |r| < 1$$

This is the series multiplied by p in our expected value calculation. The series converges if $|r| < 1$, which it does because p is a probability between 0 and 1 so $(1-p)$ is positive and less than 1.

$$\sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1-r)^2} = \frac{1}{[1-(1-p)]^2} = \frac{1}{p^2}$$

This makes $\mathbb{E}(X) = p\frac{1}{p^2} = \frac{1}{p}$.

$$\mathbb{V}(X) = \mathbb{E}(X^2) - \left[\frac{1}{p}\right]^2$$

$$= \sum_{x=1}^{\infty} x^2(1-p)^{x-1}p - \frac{1}{p^2} = p\sum_{x=1}^{\infty} x^2(1-p)^{x-1} - \frac{1}{p^2}$$

By differentiating a geometric series twice, it can be shown that the series, $\sum_{x=1}^{\infty} x^2(1-p)^{x-1} = \frac{2-p}{p^3}$. So,

$$\mathbb{V}(x) = p\frac{2-p}{p^3} - \frac{1}{p^2}$$
$$= \frac{2-p-1}{p^2} = \frac{1-p}{p^2}$$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import scipy.special as sc

# Probability grid
p_grid = np.linspace(0.001, 0.999, 500)

# 1. Continuous: Standard Normal
# Approximate by Monte Carlo
q_norm = np.quantile(np.random.normal(size=1000000), p_grid)

# Or exact using inverse error function
q_norm = np.sqrt(2) * sc.erfinv(2 * p_grid - 1)

# 2. Discrete: W = number of Heads minus number of Tails (3 fair coin flips)
values = np.array([-3, -1, 1, 3])
probs  = np.array([1.0/8.0, 3.0/8.0, 3.0/8.0, 1.0/8.0])
cum_probs = np.cumsum(probs)

# Step-like quantile function for discrete distribution
def discrete_quantile(p, x_vals, cum_p):
    # Find the smallest index where cum_p is greater than p
    idx = np.searchsorted(cum_p, p, side="right")
    idx = np.clip(idx, 0, len(x_vals) - 1)
    return x_vals[idx]

q_disc = discrete_quantile(p_grid, values, cum_probs)

# Plotting
plt.figure(figsize=(12, 5))

# Left: Normal
plt.subplot(1, 2, 1)
plt.plot(p_grid, q_norm, lw=2.2)
plt.title("Quantile Function (Inverse CDF)\nStandard Normal Distribution", fontsize=13, pad=10)
plt.xlabel("q", fontsize=12)
plt.ylabel("F_inverse(q)", fontsize=13)
plt.grid(alpha=0.15, linestyle="--")
```

```
plt.minorticks_on()

# Right: Discrete (3 coin flips)
plt.subplot(1, 2, 2)
plt.step(p_grid, q_disc, where="pre", lw=2.2, color="darkorange")

# Show jump points
plt.plot(cum_probs, values, "o", ms=8, mec="navy", mfc="white",
         mew=1.5, zorder=5, label="Probability jumps")

# Vertical dashed lines at jump locations
for cp in cum_probs[:-1]:
    plt.axvline(cp, color="gray", ls="--", alpha=0.25, lw=1)

plt.title("Quantile Function (Inverse CDF)\nDiscrete: W = Heads minus Tails (3 flips)",
          fontsize=13, pad=10)
plt.xlabel("q", fontsize=12)
plt.ylabel("F_inverse(q)", fontsize=13)
plt.yticks([-3, -1, 1, 3])
```

```
## ([<matplotlib.axis.YTick object at 0x000002B7D114EFE0>, <matplotlib.axis.YTick object at 0x000002B7D
```

```
plt.grid(alpha=0.15, linestyle="--")
plt.minorticks_on()
plt.legend(loc="upper left", fontsize=10, framealpha=0.92)

plt.tight_layout()
plt.show()
```